

Design Document: Mach-Zehnder Interferometer

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Abstract: The primary objective of this design report is to fabricate and test an unbalanced Mach-Zehnder Interferometer (MZI) circuit to experimentally extract the waveguide group index (n_g) near the 1550 nm wavelength. By testing six MZI variations with incrementally increasing imbalance lengths (ΔL), we will observe the trend of the Free Spectral Range (FSR) varying with ΔL and compare the experimentally extracted n_g against Lumerical FDTD and MODE simulations.

1. Introduction

Silicon photonics provides a scalable platform for dense photonic integrated circuits across optical communications, sensing, and computing applications. The fundamental routing backbone of these circuits relies on high-index-contrast silicon waveguides to efficiently confine and propagate light, while fiber grating couplers provide the essential vertical optical interface to couple light between off-chip single-mode fibers and sub-micron on-chip devices.

The primary design objective of this project is to construct an unbalanced Mach-Zehnder Interferometer (MZI) circuit using standard strip waveguides and TE grating couplers to experimentally extract the waveguide group index (n_g) near the 1550 nm operating wavelength. By varying the physical path length difference (ΔL) across six distinct designs, we will systematically measure the relationship between the Free Spectral Range (FSR) and ΔL , validate our Lumerical simulation models, and establish a baseline for manufacturing variability.

2. Theory

2.1 Interferometer Theory and Transfer Function

The unbalanced MZI splits an incoming optical wave of power I_{in} equally into two arms using a 3dB 50/50 splitter (ideal Y-branch). The physical path length difference between the two arms is $\Delta L = L1 - L2$. Recombining the signals at an identical Y-branch produces interference governed by the phase difference $\Delta\phi(\lambda)$:

$$\Delta\phi(\lambda) = 2\pi \frac{n_{eff}(\lambda)}{\lambda} \Delta L$$

Assuming lossless propagation and ideal 50/50 Y-branches, the optical power transfer function $T(\lambda) = \frac{I_{out}(\lambda)}{I_{in}(\lambda)}$ is given by:

$$T(\lambda) = \cos^2\left(\frac{\Delta\phi(\lambda)}{2}\right) = \frac{1}{2} \left[1 + \cos\left(2\pi \frac{n_{eff}(\lambda)\Delta L}{\lambda}\right) \right]$$

When waveguide propagation loss α (in m^{-1}) is included, the power transfer function becomes:

$$T(\lambda) = \frac{1}{4} e^{-\alpha L_1} \left[1 + e^{-\alpha \Delta L} + 2e^{-\frac{1}{2}\alpha \Delta L} \cos\left(2\pi \frac{n_{eff}(\lambda)\Delta L}{\lambda}\right) \right]$$

2.2 Derivation of Group Index Equation from Free Spectral Range

The Free Spectral Range (FSR) is defined as the spacing between consecutive transmission maxima (or minima) in the optical spectrum. Let λ_1 and λ_2 be two adjacent peak wavelengths, where $\lambda_2 > \lambda_1$ and $FSR = (\lambda_2 - \lambda_1) = \Delta\lambda$.

The phase condition for consecutive transmission peaks corresponding to integer interference orders m and $m - 1$ is:

$$\Delta\phi(\lambda_1) = \frac{2\pi n_{eff}(\lambda_1)}{\lambda_1} \Delta L = 2m\pi$$

$$\Delta\phi(\lambda_2) = \frac{2\pi n_{eff}(\lambda_2)}{\lambda_2} \Delta L = 2(m - 1)\pi$$

Subtracting the two phase expressions yields:

$$\Delta\phi(\lambda_1) - \Delta\phi(\lambda_2) = 2\pi$$

To express this phase change in terms of dispersion, we apply a first-order Taylor series expansion of $\Delta\phi(\lambda)$ around λ_1 :

$$\Delta\phi(\lambda_2) \approx \Delta\phi(\lambda_1) + \left. \frac{d\Delta\phi}{d\lambda} \right|_{\lambda_1} (\lambda_2 - \lambda_1)$$

Evaluating the derivative of phase with respect to wavelength:

$$\frac{d\Delta\phi}{d\lambda} = \frac{d}{d\lambda} \left(\frac{2\pi n_{eff}(\lambda) \Delta L}{\lambda} \right) = 2\pi \Delta L \left(\frac{\frac{dn_{eff}}{d\lambda} \cdot \lambda - n_{eff}(\lambda)}{\lambda^2} \right) = -\frac{2\pi \Delta L}{\lambda^2} \left(n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \right)$$

Recalling the definition of waveguide group index $n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda}$:

$$\frac{d\Delta\phi}{d\lambda} = -\frac{2\pi \Delta L \cdot n_g(\lambda)}{\lambda^2}$$

Substituting this derivative into the Taylor expansion and setting $|\Delta\phi(\lambda_2) - \Delta\phi(\lambda_1)| = 2\pi$:

$$\frac{2\pi \Delta L \cdot n_g}{\lambda_0^2} \Delta\lambda = 2\pi$$

Solving for the group index n_g :

$$n_g = \frac{\lambda_1 \lambda_2}{\Delta L \cdot FSR} \approx \frac{\lambda_0^2}{\Delta L \cdot FSR}$$

Where $\lambda_0 = \sqrt{\lambda_1 \lambda_2} \approx \frac{\lambda_1 + \lambda_2}{2}$ is the center wavelength between adjacent peaks.

3. Model and Simulation

3.1 Waveguide Geometry and Mode Profile

3.1.1 Waveguide Parameters

- **Core Material:** Silicon (Si , $n \approx 3.47$ at 1550 nm)
- **Cladding Material:** Silicon Dioxide (SiO_2 , $n \approx 1.448$ at 1550 nm)

- **Waveguide Cross-Section:** Width $w = 500nm$, Height $h = 220nm$
- **Polarization:** Fundamental TE (TE_{00})

3.1.2 Mode Profile Simulation

The modal distribution of the fundamental TE mode was simulated in Lumerical MODE Solutions using the Finite Difference Eigenmode (FDE) solver. The fundamental TE mode exhibits strong dominant transverse electric field polarization E_x , with the optical energy tightly confined within the $500\text{ nm} \times 220\text{ nm}$ silicon core.

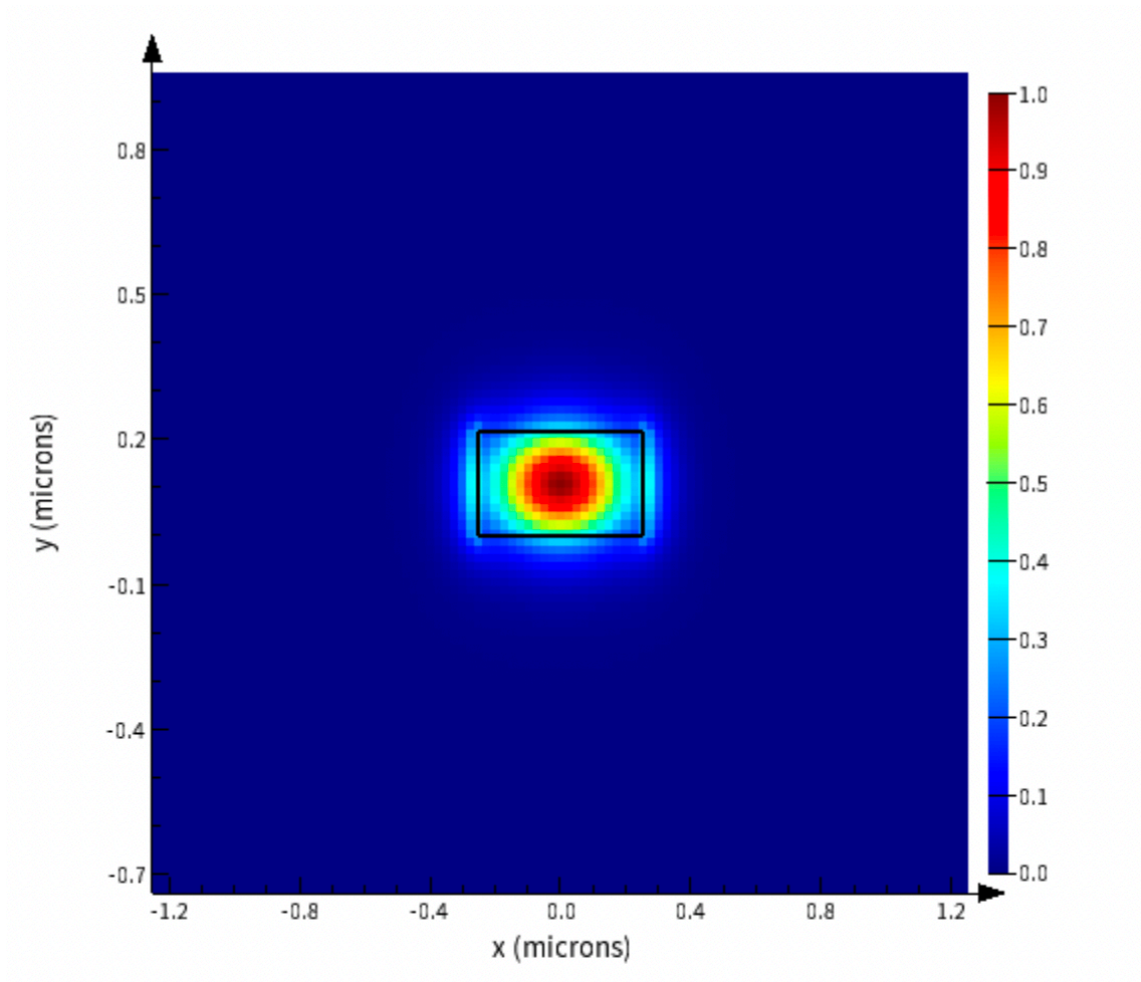


Figure 1: Lumerical Mode FDE Cross-Sectional Field Profile for a 500 nm x220 nm silicon strip waveguide at $\lambda = 1550\text{ nm}$

3.2 Compact Model and Waveguide Dispersion

3.2.1 Effective Index (n_{eff}) and Group Index (n_g) Curves

Using the frequency sweep feature in Lumerical MODE over $1.5 \mu m$ to $1.6 \mu m$ wavelength range, the wavelength dependence of the effective index (n_{eff}) and group index (n_g) was calculated.

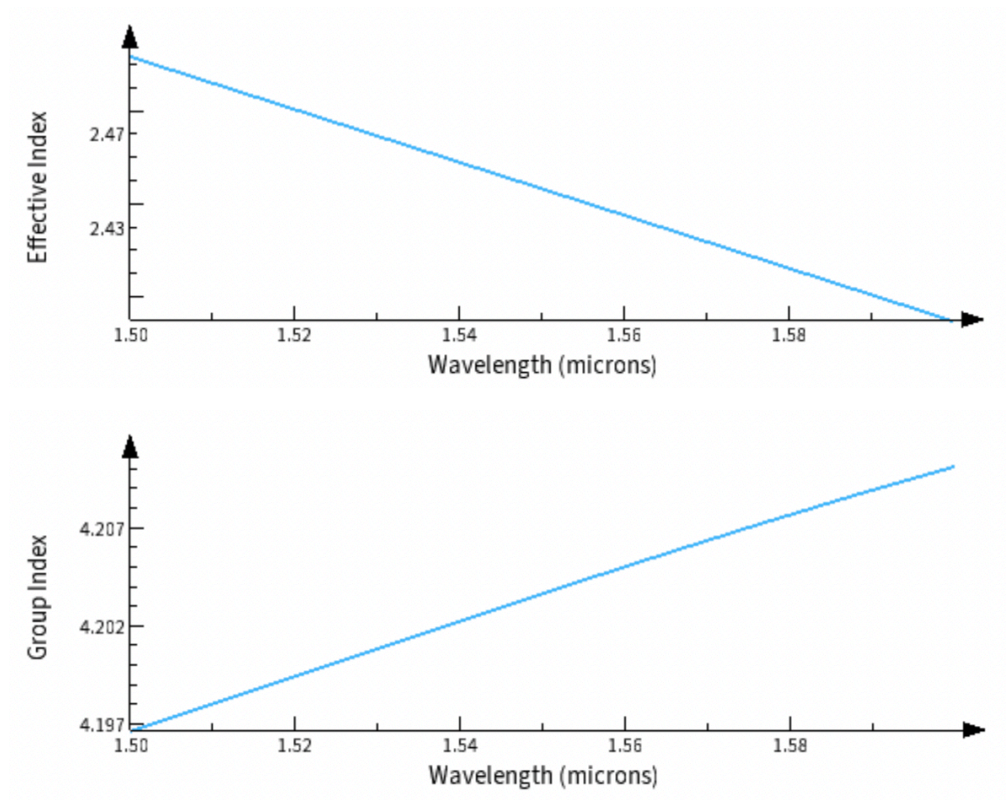


Figure 2: Simulated Effective Index (n_{eff}) and Group Index (n_g) versus wavelength ($\lambda = 1.5 \mu m$ to $1.6 \mu m$) for Fundamental TE mode

At the target wavelength $\lambda_0 = 1550 \text{ nm}$, the nominal values determined from simulation are:

- Effective Index (n_{eff}): $n \approx 2.44$
- Group Index (n_g): $n_g \approx 4.20$

3.2.2 Compact Model Polynomial Expression

To enable fast circuit-level simulations in Lumerical INTERCONNECT, $n_{eff}(\lambda)$ is expanded as a second-order polynomial centered at $\lambda_0 = 1.55 \mu m$:

$$n_{eff}(\lambda) = n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2$$

Where λ is specified in micrometers (μm), and the polynomial coefficients obtained from the Lumerical polynomial fit script are:

$$n_0 = n_{eff}(\lambda_0) \approx 2.4412$$

$$n_1 = \left. \frac{dn_{eff}}{d\lambda} \right|_{\lambda_0} \approx -1.1341 \mu\text{m}^{-1}$$

$$n_2 = \frac{1}{2} \left. \frac{d^2 n_{eff}}{d\lambda^2} \right|_{\lambda_0} \approx 0.1250 \mu\text{m}^{-2}$$

The group index $n_g(\lambda)$ is related to the effective index by:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda}$$

3.3. Extract Group Index

3.3.1 Group Index Through Simulation.

Using the baseline simulation parameters ($\Delta L = 40 \mu\text{m}$, $\lambda_0 = 1.55 \mu\text{m}$, simulated $n_g = 4.20$):

$$\text{FSR}_{sim} = \frac{(1.55 \times 10^{-6})^2}{4.20 \times (40 \times 10^{-6})} = 1.4315 \times 10^{-8} \text{ m} = 14.315 \text{ nm}$$

Re-extracting n_g from FSR_{sim} :

$$n_g = \frac{(1.55 \times 10^{-6})^2}{(40 \times 10^{-6}) \times (14.315 \times 10^{-9})} = 4.20$$

This confirms the mathematical consistency of the extraction formula.

3.3.2 Experimental Extraction Methodology

- Peak Detection: Measure the transmission spectrum $T(\lambda)$ across 1500 nm – 1600 nm. Find all peak wavelengths λ_m
- FSR Calculation: Compute local peak-to-peak spacing $\text{FSR}_m = \lambda_{m+1} - \lambda_m$
- Linear Regression: Plot interference peak order m versus optical frequency $\nu_m = c/\lambda_m$. The slope of $m(\nu)$ equals $\frac{n_g \Delta L}{c}$, allowing n_g to be extracted via linear regression over the full spectral band to minimize experimental noise.

3.4 Parameter Variations and Expected Performance

To isolate ΔL as the single variable, six MZI layout configurations are designed using uniform 500 nm $\times$ 220 nm TE strip waveguides, Y-branch splitters, and bend radii at 5 μm to ensure negligible bend loss ($< 0.1\%$).

All target FSR values strictly satisfy the condition of remaining within the ~ 50 nm measurement bandwidth of the optical test system.

Design	Imbalance Length (ΔL)	Waveguide Width	Waveguide Height	Polarization	Target Wavelength(λ_0)	Expected FSR (nm)
MZI 1	$25 \mu m$	500 nm	220 nm	TE	1550 nm	22.88 nm
MZI 2	$50 \mu m$	500 nm	220 nm	TE	1550 nm	11.44 nm
MZI 3	$75 \mu m$	500 nm	220 nm	TE	1550 nm	7.63 nm
MZI 4	$100 \mu m$	500 nm	220 nm	TE	1550 nm	5.72 nm
MZI 5	$125 \mu m$	500 nm	220 nm	TE	1550 nm	4.58 nm
MZI 6	$150 \mu m$	500 nm	220 nm	TE	1550 nm	3.81 nm

Table 1. Six MZI layout configurations

3.5 Transmission Spectrum Simulations

Using the compact waveguide model inside Lumerical INTERCONNECT, circuit-level simulations were executed for the unbalanced MZI structures.

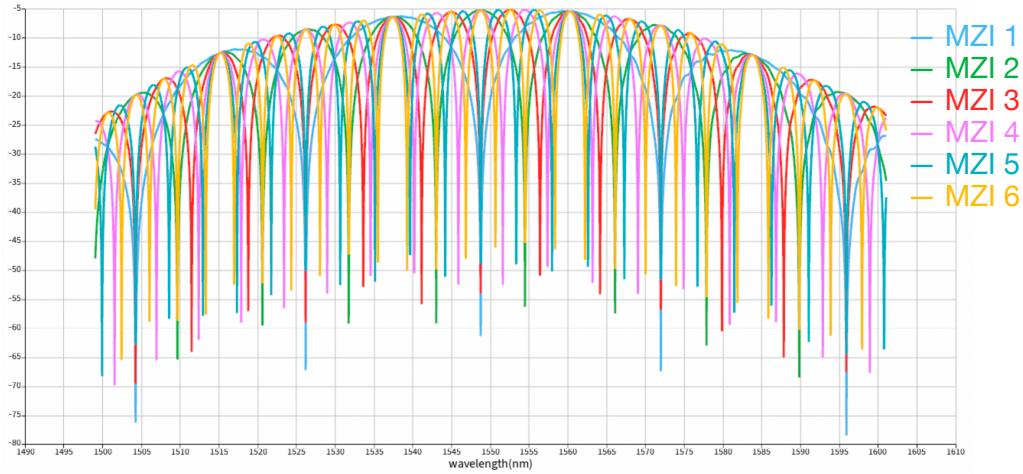


Figure 3: Simulated Transmission Spectrum (dB) vs. Wavelength for MZI Variations ($\Delta L = 25 \mu m$ to $150 \mu m$) showing interference fringes and grating coupler envelope.

The simulated transmission exhibits sinusoidal interference fringes superimposed on the Gaussian bandpass response of the input/output fiber grating couplers. As ΔL increases from $25 \mu m$ to $150 \mu m$, the fringe density increases, corresponding to the expected reduction in FSR from 22.88 nm to 3.81 nm.

4. Fabrication

(To be completed)

5. Experiment data

(To be completed)

6. Analysis

(To be completed)

7. Conclusion

(To be completed)

8. References

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2. Lukas Chrostowski, Silicon Photonics Design, Fabrication and Data Analysis EDX Course Material, 2026